



SATA SSD Module Datasheet

2259XT2

128GB~2TB



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Preliminary version changes without notice



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1. Introduction

The SATA-III SSD provides high reliability, high stability and high performance for a storage media. The drive doesn't have any moving parts such as platter (disk) and head media, which provides a better solution in a laptop or Tablet PC. It provides several advantages in higher performance, high input/output rate, reduced latencies, lower power consumption and lighter weight in a small form factor in compare to traditional hard drive.

The SATA-III SSD drive adopted SMI (Silicon Motion Inc.) controller to cover a board range of 3D NAND support from major NAND supply vendors.

It's in an easily installed standard 2.5" form factor. There is a range of capacities available from 128GB to 2TB.





2. Controller Key Feature

◆ Host Interface

- SATA 3.1 compliant
- ATA/ATAPI-8 & ACS-3 command compliant
- SATA Interface rate of 6Gb/s (backward compatible to 1.5 and 3 Gb/s)
- Native Command Queuing up to 32 commands
- Data Set Management command (TRIM support)
- Support S.M.A.R.T (Self-Monitoring, Analysis, and Reporting Technology)
- 28-bit & 48-bit LBA mode command

◆ NAND Flash interface

- ONFI v4.0 and Toggle 2.0 interface
- SDR, NV-DDR, NV-DDR2, NV-DDR3 and Toggle DDR/DDR2 NAND Flash
- 2-channel Flash interface support up to 16 NAND Flash devices
- 1.2V/1.8V Flash I/O
- 8KB and 16KB page size
- 1/2/4/8 plane operation

◆ Data Protection & Reliability

- End-to-end Protection
- ATA8 security feature set
- Hardware LDPC ECC engine with hard-decision and soft-decision decoding
- RAID engine offers additional level of data protection
- Internal data shaping technology

- Static Data Refresh technology ensures data integrity
- Early weak block retirement option
- Global wear leveling algorithm evens program/erase count and extends SSD lifespan

■ Architecture

- 32-bit RISC CPU
- High-efficiency 64-bit system bus
- Automatic sleep & wake-up mechanism to save power
- Built-in voltage detectors for power failure protection
- Built-in power-on reset and voltage regulators
- Built-in JTAG interface and UART (RS-232) interface

■ Enhanced Security

- Real time full drive encryption with AES (Advanced Encryption Standard)
- Hardware SHA 256 and TRNG (True Random Number Generator)

■ Package

- 144-ball TFBGA
- Lead-free and RoHS compliant



increase data endurance

3. Module Specification

SATA SSD w/ SMI SM2259XT2		
Interface	SATA III 6Gb/s and backwards compatible	
Form Factor	2.5" 7mm	
Disk Capacity	128GB~2TB	
Seq. Read/Write	128GB~2TB up to 530/490MB/s	
Random 4K Read/Write	128GB~2TB up to 46.5K/70K IOPS	
TBW	128GB - 75TB 256GB - 140TB 512GB - 280TB 1TB - 520TB 2TB - 1200TB	
Power Consumption	80mW Slumber / 7mW avg. 2.5W (MAX) read / 2W (MAX) write	
Storage Temp.	-40°C to 85°C	
Operating Temp.	0°C to 70°C	
Dimensions	2.5" 6.9±0.1mm	100.0mm x 69.85mm x 7.0mm
Weight	8~11g	
Shock@ Operating	0.5ms / 1500G peak	
Shock@ Non-Operating	0.5ms / 1500G peak	
Vibration@ Operating	3G peak (2-500Hz)	
Vibration@ Non-Operating	20G peak (20-2000Hz)	
MTBF	1,500,000hrs	
Warranty	limited 3-years warranty with free technical support	

* Read/Write performance is base max. performance can be measured, speed may vary due to host hardware, software and usage.

* Total Bytes Written (TBW) is derived from the JEDEC Client Workload (JESD219A)

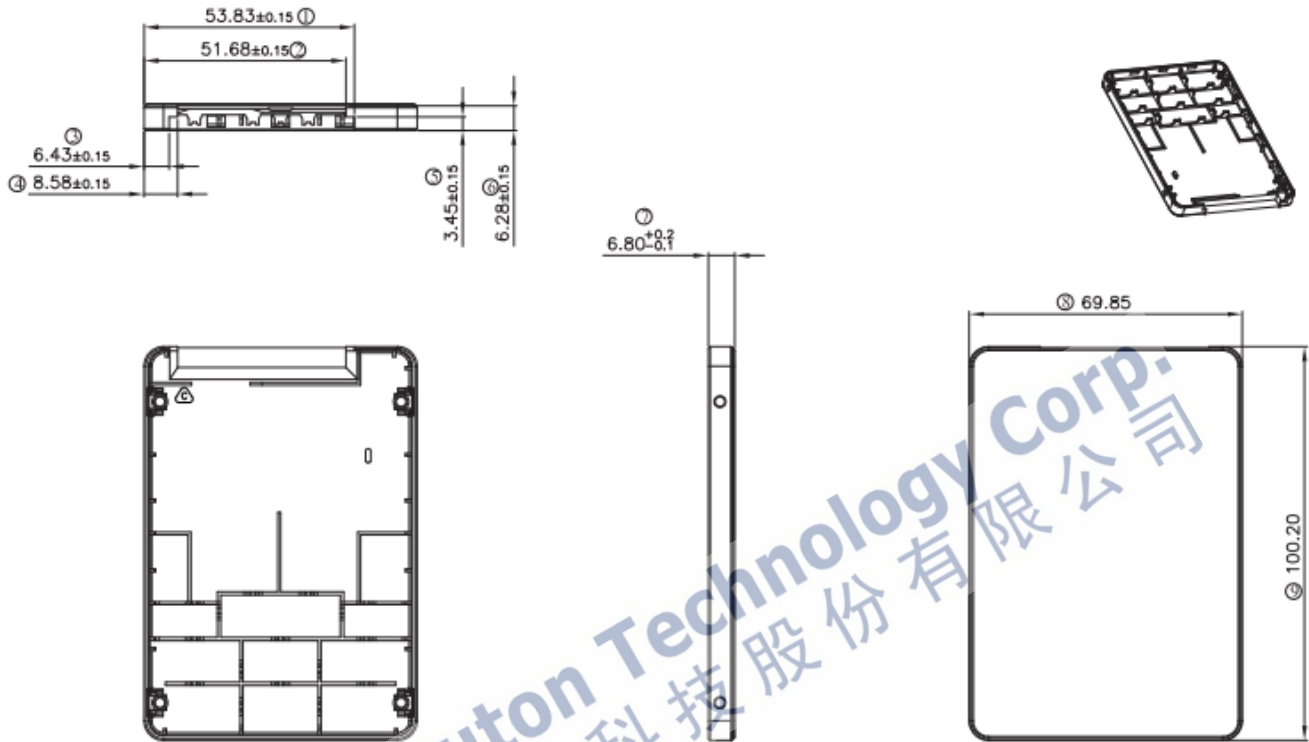
* Limited warranty based on 3-year or "Percentage Used" based on TBW limit



4. Mechanical Form Factor

Dimensions for 2.5" / 7mm

Top



Bottom

